$xRy \Leftrightarrow x^2 - y^2 = x - y$

① Ref: xRx since $x^2 - x^2 = x - x$
$= 0$ ✓

② Sym: If $xRy \Rightarrow yRx$

$xRy \Leftrightarrow x^2 - y^2 = x - y$

$\Rightarrow -(x^2 - y^2) = -(x - y)$

$\Rightarrow y^2 - x^2 = y - x \Rightarrow yRx$

③ Tran: Let $x, y, z \in \mathbb{R}$

$\begin{cases} xRy \\ yRz \end{cases} \Rightarrow \begin{matrix} x^2 - y^2 = x - y \\ y^2 - z^2 = y - z \end{matrix} \Rightarrow x^2 - z^2 = x - z \Rightarrow xRz$

Then: R is an equiv relation

$P = \{(x,y) \in \mathbb{R}^2 / xRy\}$

$= \{(x,y) \in \mathbb{R} / x^2 - y^2 = x - y\}$

$= \{(x,y) \in \mathbb{R} / (x-y)(x+y) - (x-y) = 0\}$

$(x-y)(x+y) - (x-y) = 0$

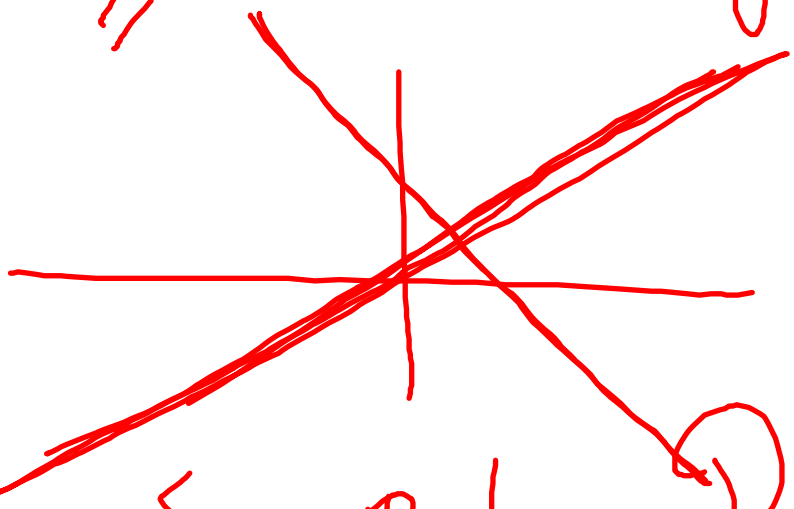
$$= \{(x,y) \in \mathbb{R}^2 / (x-y)(x+y-1)=0\}$$

$$= \{(x,y) \in \mathbb{R}^2 / \; x=y \vee x+y-1=0\}$$

$$= \Delta_1 \cup \Delta_2.$$

Δ_1 la droite d'équation $x=y$

Δ_2 " " $y = 1-x$



$\overline{O} = d(O) = \{x \in \mathbb{R} \mid x^2 \geq 0\}$

$x^2 - 0 = x - 0 \iff x^2 - x = 0$

$\Rightarrow x(x-1)=0$

$\Rightarrow \{x=0 \vee x=1\}$

$\overline{O} = \{0,1\}$

$1 \in \overline{O} \Rightarrow \overline{1} = \overline{O} = \{0,1\}$

$$\overline{\left(\frac{1}{2}\right)} = \{x \in \mathbb{R} / x\mathcal{R}\tfrac{1}{2}\} = \{\tfrac{1}{2}\}$$

$$x^2 - \frac{1}{4} = x - \tfrac{1}{2}$$

$$\left(x-\tfrac{1}{2}\right)\left(x+\tfrac{1}{2}\right) - \left(x-\tfrac{1}{2}\right) = 0$$

$$\left(x-\tfrac{1}{2}\right)\left(x+\tfrac{1}{2}-1\right) = 0$$

$$\Rightarrow \left(x-\tfrac{1}{2}\right)\left(x-\tfrac{1}{2}\right) = 0$$